

NLP - Text representation

Computers understand only numbers and vectors. As for images classification or a regression problem, NLP needs features to process text. This tutorial helps you to understand what are features for a text.

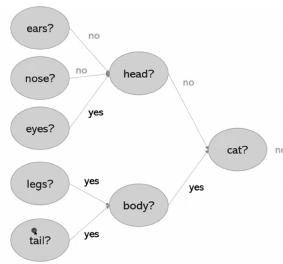


FIGURE 1 – Cat’s Features

Mbappe	S. Jobs	France
Person : 1	Person : 1	Person : 0
Healthy : 0.9	Healthy : 0.5	Healthy : 0.7
Location : 0	Location : 0	Location : 1
Has two eyes : 1	Has two eyes : 1	Has two eyes : 0
Government : 0	Government : 0	Government : 1

Mbappe	S. Jobs	France
1	1	0
0.9	0.5	0.7
0	0	1
1	1	0
0	0	1

TABLE 1 – Mbappe, S. Jobs and France vectors

Feature engineering is the process of extracting features from raw data (Text).



FIGURE 2 – Approaches to convert text into vectors

1. Label encoding is based on a given vocabulary. We assign a label to each word. It is a primitive technique. No used nowadays.

Text	Vocabulary
Hi	0
Zizou	0
are	10
you	220
happy	70
today	155
...	...

2. One-Hot encoding is based on a given vocabulary. We assign either 1 or 0 to each word and we obtain a binary vector. It is a primitive technique. No used nowadays. Indeed, it doesn't capture similarities between words and consumes too much memory. Moreover it can't process words that are Out Of Vocabulary (OOV).

Text	Vocabulary
Hi	[0 0 0 0 0 ... 0]
Zizou	[0 0 0 0 0 ... 0]
are	[0 0 0 0 0 ... 1]
you	[0 0 0 0 0 ... 0]
happy	[1 0 0 0 0 ... 0]
today	[0 1 0 0 0 ... 0]
...	...

3. Bag of Words : n-grams, $n = \text{nbre of words}$.

Ali drinks water

Pierre is heavy

Pierre is drinking water

Document	Ali	drinks	water	Pierre	heavy	"Ali drinks"	"drinks water"	"Pierre heavy"	"Pierre drink"
Doc 1	[1	1	1	0	0	1	1	0	0]
Doc 2	[0	0	0	1	1	0	0	1	0]
Doc 3	[0	1	1	1	0	0	1	0	1]

Frequencies calculation gives meanings to docs.

4. TF-IDF text representation

What is TF-IDF ?

TF stands for Term Frequency. It is the ratio of number of times a particular word appeared in a Document to total number of words in the document : so,

$$(TF) = \frac{\text{number_of_times_word_appeared}}{\text{total_no_of_words_in_a_document}}$$

Term Frequency values ranges between 0 and 1. If a word occurs more number of times, then it's value will be close to 1.

IDF stands for Inverse Document Frequency and denotes the log of ratio of total number of documents in the whole dataset to the number of documents that contains the particular word, so :

$$(IDF) = \frac{\log(\text{Total_number_of_documents})}{\text{number_of_documents_that_contains_the_word}}$$

In IDF, if a word occurred in more number of documents and is common across all documents, then it's value will be less and ratio will approaches to 0

$$TF - IDF = \text{Term_Frequency}(TF) \times \text{Inverse_Document_Frequency}(IDF)$$

5. BOW Example with Jupyter notebook, scikit-learn and spaCy :

- After searching for the CountVectorizer() function, create a vocabulary using a sentence "The Internet of Things master's degree requires natural language processing skills" e. (Hint : use .fit and .vocabulary_ functions exploring scikit-learn documentation)
- Create a (1,2) ngram range vocabulary.
- Same question with range (1,3).
- Let's consider now the corpus (i.e. a collection of documents) of the 3 documents : "The Internet of Things master's ", "master's degree requires" and "natural

language processing skills". With spaCy, define a preprocessing function that gives all the tokens that are not stopwords neither punctuations. Lemmatize them. (Use the (1,2)) ngram range vocab).

- (e) use this function to preprocess the corpus.
- (f) Create the vocabulary of the corpus.
- (g) Using *.transform* function convert the sentences of the corpus to vectors.
- (h) Same question with "master's degree STGI".

1. TF-IDF Example

- (a) Import the vectorize count vectorizer from scikit_learn
- (b) fit (create vocabulary) and vectorize the following corpus :
"Thor eating pizza, Loki is eating pizza, Ironman ate pizza already",
"Apple is announcing new iphone tomorrow",
"Tesla is announcing new model-3 tomorrow",
"Google is announcing new pixel-6 tomorrow",
"Microsoft is announcing new surface tomorrow",
"Amazon is announcing new eco-dot tomorrow",
"I am eating biryani and you are eating grapes"
- (c) print the vocabulary using TfidfVectorizer()
- (d) print the IDF of each word
- (e) Print the transformed outputs (vectors) from TF-IDF